

Jialong Wu

(+86) 13587027920 | wujialong0229@gmail.com | manchery.github.io | [Google Scholar](https://scholar.google.com/citations?user=JialongWu)

EDUCATION

Ph.D. in Software Engineering

School of Software, Tsinghua University

- Advisor: [Prof. Mingsheng Long](#)
- Research Interests: World Models, Reinforcement Learning

Aug 2022 – June 2027 (*expected*)

Beijing, China

Bachelor in Software Engineering

School of Software, Tsinghua University

- GPA: 3.98/4.0, Rank: 1/89

Aug 2018 – June 2022

Beijing, China

Bachelor in Pure and Applied Mathematics (*Second Degree*)

Department of Mathematical Sciences, Tsinghua University

- GPA: 3.75/4.0

Aug 2019 – June 2022

Beijing, China

EXPERIENCE

Research Intern

ByteDance Seed

March 2025 – *present*

Beijing, China

PUBLICATIONS AND PREPRINTS

Visual Generation Unlocks Human-Like Reasoning through Multimodal World Models

Jialong Wu, Xiaoying Zhang#, Hongyi Yuan, Xiangcheng Zhang, Tianhao Huang, Changjing He, Chaoyi Deng, Renrui Zhang, Youbin Wu, Mingsheng Long#. *arXiv preprint*, 2026.

[[Project website](#)] [[PDF](#)] [[Code](#)] [[Slides](#)] [[Datasets](#)]

Thoth: Mid-Training Bridges LLMs to Time Series Understanding

Jiafeng Lin*, Yuxuan Wang*, Jialong Wu*, Huakun Luo, Zhongyi Pei#, Jianmin Wang. *arXiv preprint*, 2026.

[[PDF](#)] [[Code](#)] [[Models](#)]

JEPA-VLA: Video Predictive Embedding is Needed for VLA Models

Shangchen Miao, Ningya Feng, Jialong Wu, Ye Lin, Xu He, Dong Li, Mingsheng Long#. *arXiv preprint*, 2026.

[[PDF](#)]

Audio-Visual World Models: Towards Multisensory Imagination in Sight and Sound

Jiahua Wang*, Shannan Yan*, Leqi Zheng*, Jialong Wu, Yaixin Mao. *arXiv preprint*, 2025.

[[PDF](#)]

Vid2World: Crafting Video Diffusion Models to Interactive World Models

Siqiao Huang*, Jialong Wu*, Qixing Zhou, Shangchen Miao, Mingsheng Long#. *ICLR*, 2026.

[[Project website](#)] [[PDF](#)] [[Code](#)]

Foresight Diffusion: Improving Sampling Consistency in Predictive Diffusion Models

Yu Zhang*, Xingzhuo Guo*, Haoran Xu, Jialong Wu, Mingsheng Long#. *ICLR*, 2026.

[[PDF](#)]

RLVR-World: Training World Models with Reinforcement Learning

Jialong Wu, Shaofeng Yin, Ningya Feng, Mingsheng Long#. *NeurIPS*, 2025.

[[Project website](#)] [[PDF](#)] [[Code](#)] [[Models & Datasets](#)] [[Poster](#)]

Seed1.5-VL Technical Report

Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, ... , Jialong Wu (*Contributor*), et al. *Technical report*, 2025.

[[Project website](#)] [[PDF](#)] [[Code](#)]

Trajectory World Models for Heterogeneous Environments

Shaofeng Yin*, Jialong Wu*, Siqiao Huang, Xingjian Su, Xu He, Jianye Hao, Mingsheng Long#. *ICML*, 2025.

[[PDF](#)] [[Code](#)]

CompilerDream: Learning a Compiler World Model for General Code Optimization

Chaoyi Deng*, **Jialong Wu***, Ningya Feng, Jianmin Wang, Mingsheng Long#. *KDD*, 2025.

[PDF] [Code]

Long-Sequence Recommendation Models Need Decoupled Embeddings

Ningya Feng*, Junwei Pan*, **Jialong Wu***, Baixu Chen, Ximei Wang,

Qian Li, Xian Hu, Jie Jiang, Mingsheng Long#. *ICLR*, 2025.

[PDF] [Code]

iVideoGPT: Interactive VideoGPTs are Scalable World Models

Jialong Wu*, Shaofeng Yin*, Ningya Feng, Xu He, Dong Li, Jianye Hao, Mingsheng Long#. *NeurIPS*, 2024.

[Project Website] [PDF] [Code] [Models] [Poster] [Slides] [Blog]

HarmonyDream: Task Harmonization Inside World Models

Haoyu Ma*, **Jialong Wu***, Ningya Feng, Chenjun Xiao, Dong Li, Jianye Hao, Jianmin Wang, Mingsheng Long#. *ICML*, 2024.

[PDF] [Code] [Poster] [Slides] [Talk] [Blog]

Pre-training Contextualized World Models with In-the-wild Videos for Reinforcement Learning

Jialong Wu*, Haoyu Ma*, Chaoyi Deng, Mingsheng Long#. *NeurIPS*, 2023.

[PDF] [Code] [Poster] [Slides] [Blog]

CLIPood: Generalizing CLIP to Out-of-Distributions

Yang Shu*, Xingzhuo Guo*, **Jialong Wu**, Ximei Wang, Jianmin Wang, Mingsheng Long#. *ICML*, 2023.

[PDF] [Code]

Supported Policy Optimization for Offline Reinforcement Learning

Jialong Wu, Haixu Wu, Zihan Qiu, Jianmin Wang, Mingsheng Long#. *NeurIPS*, 2022.

[PDF] [Code] [Poster] [Slides]

Out-of-Dynamics Imitation Learning from Multimodal Demonstrations

Yiwen Qiu, **Jialong Wu**, Zhangjie Cao, Mingsheng Long#. *CoRL*, 2022.

[PDF] [Code]

Flowformer: Linearizing Transformers with Conservation Flows

Haixu Wu, **Jialong Wu**, Jiehui Xu, Jianmin Wang, Mingsheng Long#. *ICML*, 2022.

[PDF] [Code]

HONORS AND AWARDS

ByteDance Scholarship (awarded to 20 students in China and Singapore)	2025
ByteDance [Certificate]	
ICLR 2025 Notable Reviewer	2025
ICLR 2025 [Link]	
National Scholarship (Top 2%)	2020, 2024
Ministry of Education of the People's Republic of China [Certificate 2020] [Certificate 2024]	
Excellent Teaching Assistant Award	2024
Tsinghua University [Certificate]	
NeurIPS 2023 Top Reviewer	2023
NeurIPS 2023 [Link] [Certificate]	
Future Researcher Scholarship	2022
Tsinghua University [Certificate]	
Outstanding Graduates (Top 2%)	2022
School of Software [Certificate] & Tsinghua University [Certificate] & Beijing [Certificate]	
Excellent Graduation Thesis	2022
Tsinghua University [Certificate]	

Comprehensive Excellence Award (Top 10%)

2019, 2021, 2023, 2025

Tsinghua University [[Certificate 2019](#)] [[Certificate 2021](#)] [[Certificate 2023](#)]**34th National Olympiad in Informatics (NOI 2017), Silver Medal (81st Place)**

July 2017

China Computer Federation (CCF) [[Certificate](#)] [[Link](#)]INVITED TALKS AND ACTIVITIES

A Path Towards General World Models

June, 2025

[BAAI Conference](#), Beijing, China**Panelist**

Feb, 2025

[AIFC](#) Side Event of the [AI Action Summit 2025](#), Paris, France**Towards General World Models: Pre-training, Multi-Modality, and Scalable Architecture**

June, 2024

ByteDance Research

Intrinsic Task Harmonization Inside World Models

Feb, 2024

[RLChina](#)PROFESSIONAL SERVICES

Reviewer | PC Member

- International Conference on Learning Representations (*ICLR*) 2024 – 2026
- Conference on Neural Information Processing Systems (*NeurIPS*) 2023 – 2025
- International Conference on Machine Learning (*ICML*) 2022 – 2026
- Conference on Computer Vision and Pattern Recognition (*CVPR*) 2026
- Transactions on Machine Learning Research (*TMLR*)

Teaching Assistant

- Deep Learning, Fall 2023, Tsinghua University. Instructor: [Prof. Mingsheng Long](#).
- Introduction to Algorithms, Spring 2023, Tsinghua University. Instructor: [Prof. Bin Wang](#).
- Machine Learning, Fall 2022, Tsinghua University. Instructor: [Prof. Mingsheng Long](#).
- Introduction to Artificial Intelligence, Spring 2021, 2022, 2024, Tsinghua University. Instructor: [Prof. Mingsheng Long](#).

Student Counselor

- Undergraduate Student Counselor, [Dushi College](#), Tsinghua University, 2024–2027 (*expected*).

SELECTED PROJECTS

MiniVeo3-Reasoner: Thinking with Videos from Open-Source Priors | 170+ Stars

2025

We reproduce chain-of-frames visual reasoning by fine-tuning open-source video models.

[Awesome Multi-task Learning](#) | 760+ Stars

Up-to-date

A curated list of datasets, codebases, and papers on Multi-Task Learning (MTL).

ImageLabeler | C++, Qt | 120+ Stars, 40+ Forks

Summer 2019

Image annotation tool, supporting 2D/3D detection/segmentation annotation.

[Incremental Decision Tree](#) | Python (NumPy, Matplotlib, Pandas)

Fall 2020

Re-implementation of VFDT and EFDT, reproducing paper *Extremely Fast Decision Tree*.TECHNICAL SKILLS

Programming Language: Python, C/C++, MATLAB, Java, HTML/CSS, JavaScript**Professional Software:** PyTorch, JAX, NumPy, Git, LaTeX, Vue**Language:** China (native), English